Higher National Diploma in Engineering (Mechanical/Electrical)
Forth Year, First Semester Examination - 2021
ME 3227 – Industrial Engineering Management

Instructions for Candidates:
Answer any 5 questions.

No. of questions : 06
No. of pages : 03
Time : 03 hours

1)

a) Briefly explain the concept of quality. (03 marks)

b) Briefly explain any three of the following terms. (06 marks)

i) Quality control
ii) Quality improvement
iii) Quality control tools
iv) Total Quality Management

c) Discuss the key functions (activities) in human resource management. (06 marks)

d) What are the most important qualities of a leader? (05 marks)

2)

a) Explain what operations research is and why it is needed. (04 marks)

b) Write down 2 examples for OR in action. (04 marks)

A Sweet Factory produces two packages of sweets A and B. Resources used are flour, sugar, a mixture of cream and spice and man hours. Resource requirements are tabulated below.

Type of sweet package	Resource Requirement per Package of Sweet			
	Flour (Kg)	Sugar (Kg)	Cream and Spice mixture (Kg)	Man Hours
A	2.5	1.5	1	1
B	2.5	2.5	2	1

Flour and sugar are available in plenty. There are only 300Kg of cream and spice mixture and 200 man Hrs available for daily production. Maximum demand for packages of A is 150 per day. Demand for packages of B is unlimited. Profit for a package of A is Rs. 500/= and for a package of B it is Rs. 800/=.

c) Formulate this problem in linear programming. (04 marks)

d) Solve problem to find out production quantities for packages of A and B in order to obtain maximum profit. (08 marks)

3)

a) Write down the objectives of a good facilities layout. (04 marks)

b) Briefly explain the fixed position layout and group layout. (04 marks)

The Model Z torches are assembled in an assembly line. 1000 torches are required each day and available production time per day is 500 minutes. The required tasks and their performance times are given as follows.

Task	Task Time (in seconds)	Predecessor task
A	23	—
B	15	A
C	15	B
D	30	—
E	22	D,B
F	12	C,G
G	12	C,H
H	14	E
I	14	E
J	16	F,I
K	17	J
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c) Draw the network diagram. (05 marks)

d) Calculate the cycle time. (02 marks)

e) Determine the minimum no. of work stations and assign the tasks for each and every work station. (05 marks)

4)

a) What is Project Management? (02 marks)

b) Why Project Management is important? (04 marks)

c) Briefly explain project planning tools. (04 marks)

d) Explain the factors affecting for organizational design. (05 marks)

e) What are the different types of organizational structures commonly observed? (05 marks)

5)

a) Briefly explain the following terms. (4x 02 marks = 08 marks)

i) IRR ii) Profitability index

iii) Net Present Value (NPV) iv) payback period

A company is about to undertake a new project which requires to invest in a specialized machine costing Rs. 65 million. The project will last for 4 years after which the machine can be sold for Rs. 4 million. It is necessary to inject money in to the project at the end of each year to ensure that the machine is kept in proper working condition. The amount needed in 1st year is estimated to be Rs. 1.5 million. This will increase by 10% per year with respect to the previous year. The revenue produced from the project through this machine is estimated to be Rs. 20 million in 1st year and will increase by 15% per annum with respect to the previous year.

b) Tabulate the cash flows of the project. (04 marks)

c) Calculate the NPV of the project at 10% rate of discount. (04 marks)

d) Find the IRR of the project. (04 marks)

6) Briefly explain any 04 of the following topics. (Marks will be given to definitions, classifications and applications with examples) (4x 05 marks= 20 marks)

a) Life cycle of a project

b) Motivation of employees

c) e-Business

d) Determination of market price in perfectly competitive markets

e) Different Market structures

f) Importance of Industrial Psychology